

Rafting Simulator

- Realistic VR whitewater rafting experience
with immersive river navigation -

User Story Map

1. User: VR Player(beginner or rafting enthusiast)

The user of the application is a person who wants to experience the sensation of rafting in a safe and controlled environment, either to learn the basics before trying the real sport, or to enjoy the experience when they cannot practice rafting in real life.

a. Goal 1. To experience the realistic sensation of river rafting

i. Activity 1: Choosing their raft and mode

1.Feature 2: Choose between Easy Mode (simple paddle touch) or Realistic Mode (with fast-events)

2.Feature 3: Choose game mode (Free Roam - relaxed exploration, or Time Trial - race against the clock with checkpoints)

ii. Activity 2: Rafting down the river

1.Feature 1: Realistic water physics

2.Feature 2: Realistic movement of the raft in relation with the flowing water

3.Feature 3: Force coming from the paddles

4.Feature 4: Various obstacles that exist in rivers (boulders, tree trunks etc.)

5.Feature 5: Beautiful scenery to explore

6.Feature 6: Calm crystalline water sections alternating with treacherous portions

iii. Activity 3: Paddling (Easy Mode)

1.Feature 1: Touch the paddle to the water in order to move forward

2.Feature 2: Simple directional control

iv. Activity 4: Paddling (Realistic Mode with fast-events)

1.Feature 1: Quick Paddle Left/Right - make quick motions to avoid obstacles

2.Feature 2: Power Stroke Forward - strong forward paddle to push through rapids

3.Feature 3: Brace Position - prepare for impact with rocks

4.Feature 4: Hold Tight - maintain grip during small waterfalls or drops

5.Feature 5: Falling off the boat if you miss a set number/percentage of fast-events

v. Activity 5: Managing the paddle

- 1.Feature 1: Having to hold onto the paddle at all times with at least one controller**
- 2.Feature 2: Paddle will fall out of your hands if not held**
- 3.Feature 3: Pick up the paddle if it falls**
- 4.Feature 4: Get out of the raft and get the paddle if it falls in the water**
- 5.Feature 5: Swimming mechanics to retrieve paddle**

b. Goal 2: To complete challenges and improve their performance

i. Activity 1: Racing in Time Trial Mode

- 1.Feature 1: Race against the clock to reach the finish line**
- 2.Feature 2: Pass through checkpoints along the river**
- 3.Feature 3: View current time and checkpoint progress**

ii. Activity 2: Tracking performance

- 1.Feature 1: Score based on successful fast-events completed**
- 2.Feature 2: Track number of falls/mistakes**
- 3.Feature 3: Record best completion time**
- 4.Feature 4: Compare scores on global leaderboards**

c. Goal 3: To experience realistic physical feedback and immersion

i. Activity 1: Feeling physical responses

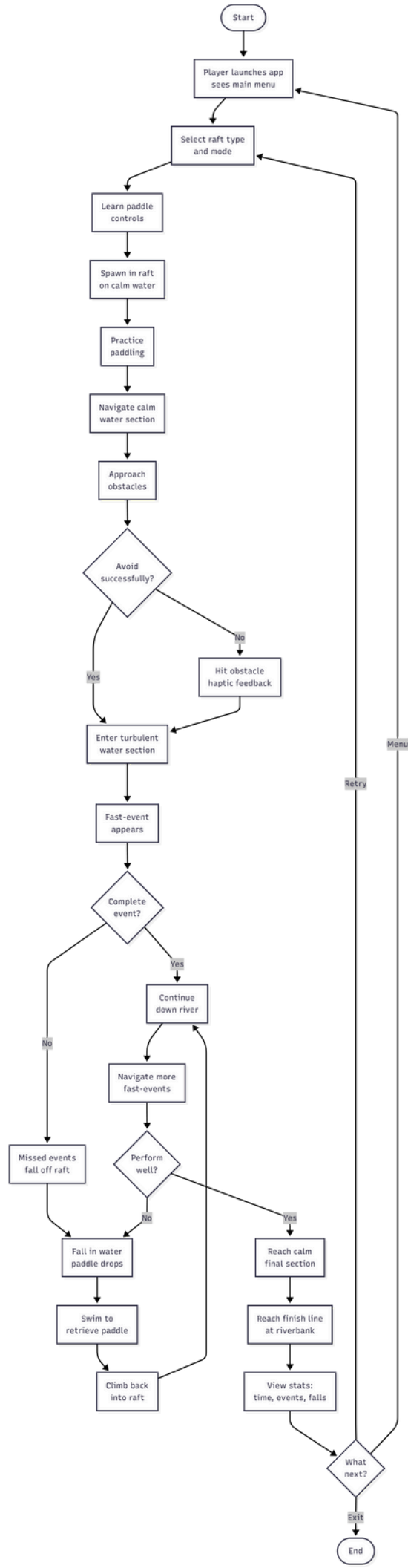
- 1.Feature 1: Haptic response when the raft hits a rock**
- 2.Feature 2: Haptic response when the player falls in the water**
- 3.Feature 3: Haptic feedback from paddle striking water**
- 4.Feature 4: Controller vibration during turbulent water sections**

ii. Activity 2: Interacting with water

- 1.Feature 1: Swimming mechanics when fallen in water**
- 2.Feature 2: Realistic water splashes and spray**
- 3.Feature 3: Visual and audio feedback from water interactions**

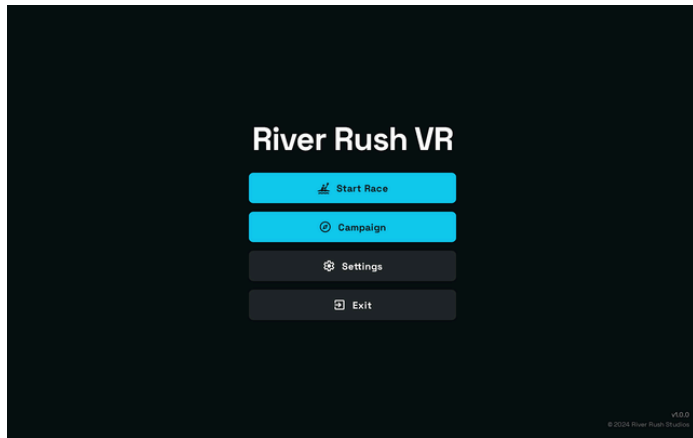
Journey Map

Player in Realistic Mode with fast-events

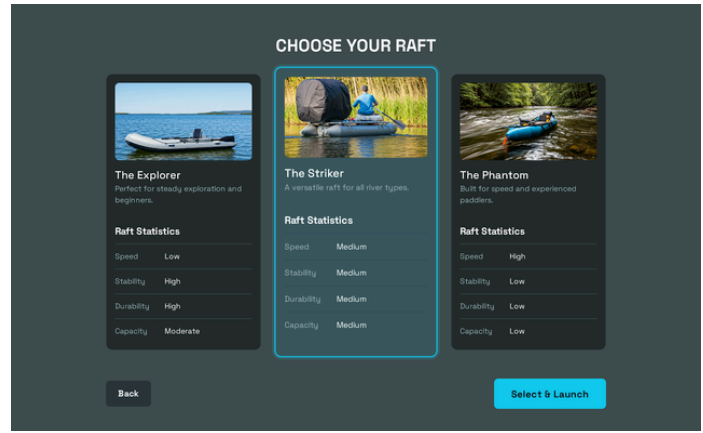


Wireframing/Storyboarding

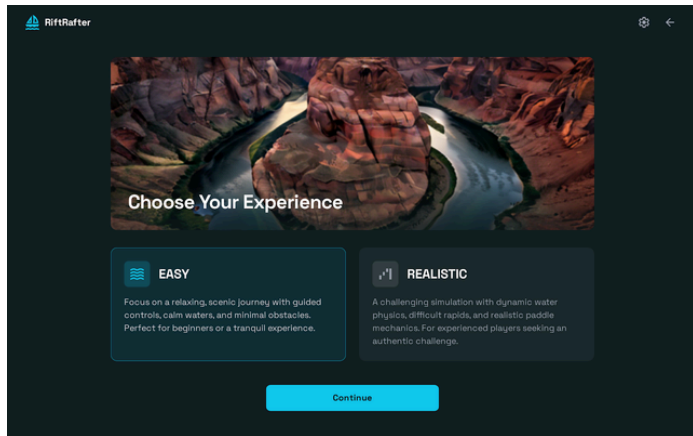
Main Menu



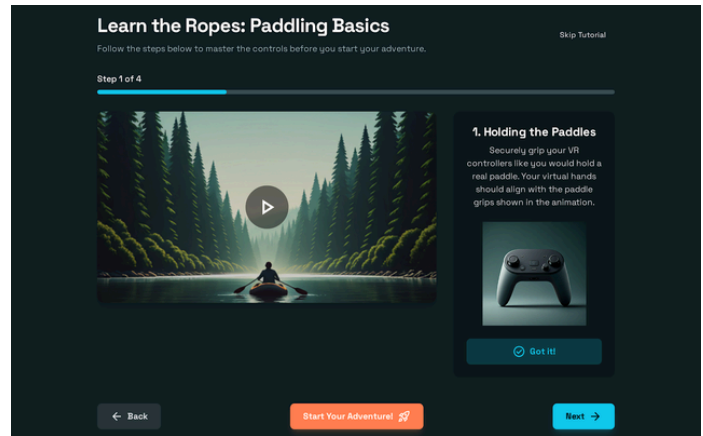
Raft Selection



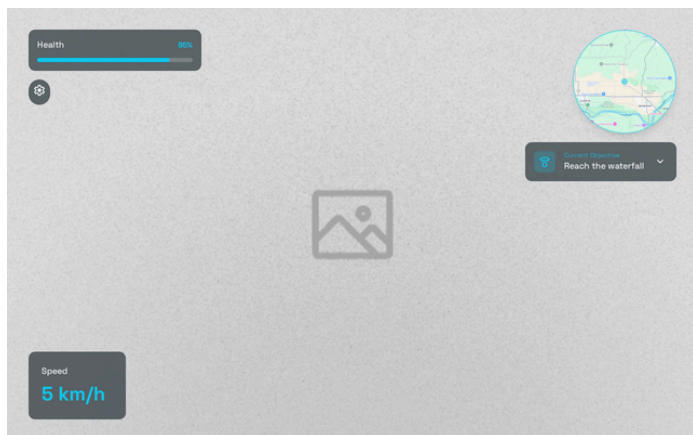
Mode Selection



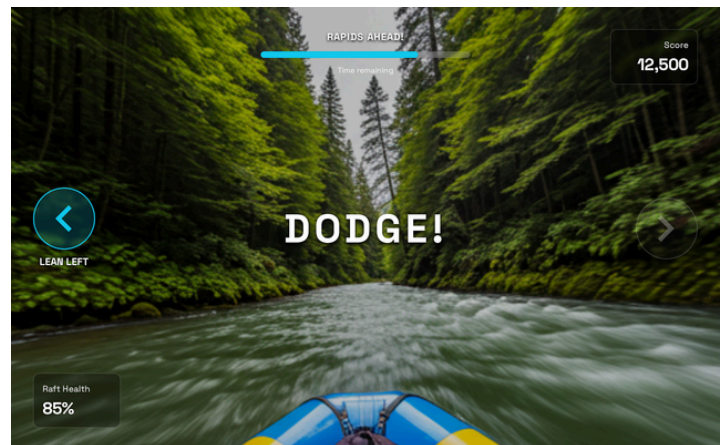
Tutorial: Paddle Controls



Calm Water Gameplay



Fast-Event Notification



Planning Asset Creation(3p)

Enumerati componentelele de care veti avea nevoie in cadrul aplicatiei, astfel:

- Subliniati(CTRL+U) materialele pe care planuiti sa le creati de la zero (modelare, scanare 3D, inregistrare sunete etc)
- **folositi un highlight galben** pentru obiectele pe care le veti genera (scanare 3D, genAI tools etc)
- scrieti sursele de unde planuiti sa le luati pe celelalte (links)

Fiți cât se poate de detaliați, această listă reprezentând practic un checklist pentru voi.

3D

(UnityAssetStore, fab.com, sketchfab, TurboSquid, CGTrader...)

➔ Raft

➔ **Paddle**

➔ River(<https://www.cgtrader.com/free-3d-models/exterior/landscape/river-062739f7-f6cf-4bc9-9961-77ed75c7ee45>)

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➔ Rocks and Boulders(<https://sketchfab.com/3d-models/river-rocks-8e10119e576f498fa5113139142e6949>)

➔ Tree Trunks (<https://assetstore.unity.com/packages/3d/vegetation/trees/polygon-trees-224068>)

➔ Checkpoints/Flags

➔ Scenery Objects (<https://www.cgtrader.com/free-3d-models/plant/leaf/forest-pack-aaa9f1f2-688d-4e53-92e0-db89d1a81637>)

➔ Wildlife (<https://assetstore.unity.com/packages/3d/characters/animals/low-poly-animals-3d-assets-258476>)

➔ Player Hands

➔ Swimming/Falling(<https://www.mixamo.com/#/?page=1&query=swimming&type=Motion%2CMotionPack> / <https://www.mixamo.com/#/?page=1&query=falling&type=Motion%2CMotionPack>)

Graphics & 2D

➔ UI ecran

➔ Avatar personaj

➔ Icon set(FlatIcon)

➔ Game logo

Sounds

(freesound.org, elevenlabs, ...)

➔ Muzica fundal-ambient river loop(Freesound
<https://freesound.org/people/Arctura/sounds/39831/>)

➔ Paddle splash sound (inregistrat)

➔ Collision impact(rocks) Mixkit

➔ Victory/Finish sound Mixkit <https://mixkit.co/free-sound-effects/discover/finish/> /
<https://mixkit.co/free-sound-effects/discover/victory/>

➔ Voice cues(ElevenLabs AI)

VFX

➔ Splash/ Water spray

➔ Foam & wave crest
(shader(<https://assetstore.unity.com/packages/3d/environments/waterworks-simple-water-ocean-river-system-for-urp-reflection-re-206909>))

➔ Sunlight reflection & glare shader (Shader Graph Ai Assistant)